

Chapter: 5

11) $\log_3 81$

$$= \log_3 3^4$$

$$= 4 \log_3 3$$

$$= 4(1)$$

$$= 4$$

12) $\log_9 81$

$$= \log_9 9^2$$

$$= 2 \log_9 9$$

$$= 2(1)$$

$$= 2$$

13) $\log_2 \left(\frac{1}{16} \right)$

$$\therefore \log_2 \left(\frac{1}{2^4} \right)$$

$$\therefore \log_2 2^{-4}$$

$$\therefore -4 \log_2 2$$

$$\therefore -4(1)$$

$$\therefore -4$$

14) $\log_5 \sqrt{625}$

$$\therefore \log_5 5^4$$

$$\therefore 4 \log_5 5$$

$$\therefore 4(1)$$

$$\therefore 4$$

15) $\log_3 81 - \log_3 9$

$$\therefore \log_3 3^4 - \log_3 3^2$$

$$\therefore 4 \log_3 3 - 2 \log_3 3$$

$$\therefore 4(1) - 2(1)$$

$$\therefore 4 - 2$$

$$\therefore 2$$

[6] $\log_4 3 \times \log_3 16$

$\Rightarrow$ use the identity $\log_a b \cdot \log_b a = 1$ (This follows from change of base)

So $\log_4 3 \times \log_3 16 = \log_4 16$

$= \log_4 4^2$

$= 2 \log_4 4$

$= 2(1)$

$= 2$

[7] $\frac{1}{\log_3 15} + \frac{1}{\log_5 15}$

$\Rightarrow$ use (reciprocal / change of - base)

use $\frac{1}{\log_a b} = \log_b a$

So

$\frac{1}{\log_3 15} + \frac{1}{\log_5 15} = \log_{15} 3 + \log_{15} 5$

$= \log_{15} (3 \times 5)$ (by Property)

$\log_a (xy) = \log_a x + \log_a y$

$= \log_{15} 15$

$= 1$

Q-2 Find Value of x / Solve the equation

11 $\log_x 16 = 4$

$$16 = x^4$$

$$x^4 = 16$$

$$x^4 = 2^4$$

$$\therefore \boxed{x = 2}$$

12 $\sqrt{\log_3 x} = 2$

$$\log_3 x = 2^2$$

$$\log_3 x = 4$$

$$x = 3^4$$

$$\therefore \boxed{x = 81}$$

13 $\log_3 (x+2) = 2$

$$\therefore (x+2) = 3^2$$

$$x+2 = 9$$

$$\therefore x = 9 - 2$$

$$\therefore \boxed{x = 7}$$

14 $\log_7 x = 1$

$$\therefore x = 7^1$$

$$\therefore x = 7$$

15 ~~$\log_7 16 = \frac{\log 64}{\log 512}$~~

$$\boxed{15} \Rightarrow \frac{\log 16}{\log x} = \frac{\log 64}{\log 512}$$

L.H.S

R.H.S

 $\Rightarrow$ First we solve R.H.S

$$\frac{\log 64}{\log 512} = \frac{\log 2^6}{\log 2^9}$$

$$= \frac{6 \log 2}{9 \log 2}$$

$$= \frac{6}{9} \frac{3 \times 0.2}{3 \times 3}$$

$$= \frac{2}{3}$$

Now L.H.S = R.H.S

So

$$\frac{\log 16}{\log x} = \frac{2}{3}$$

$$\log x = \frac{\log 16}{2/3}$$

$$= \frac{\log 2^4}{2/3}$$

$$= \frac{4 \log 2}{2/3}$$

$$= \frac{(4 \times 3) \log 2}{2}$$

$$\log x = 6 \log 2$$

$$\log x = \log 2^6$$

$$\log x = \log 64$$

$$x = 64$$

$$16) \frac{6(\log 3)(\log x)}{\log 512} = \log 81$$

$$\Rightarrow 6(\log 3)(\log x) = (\log 81)(\log 512)$$

$$6(\log 3)(\log x) = \log 3^4 (\log 2^9)$$

$$= 4 \log 3 \cdot 9 \log 2$$

$$6(\log 3)(\log x) = (4 \times 9) \log 3 \cdot \log 2$$

$$6(\log 3)(\log x) = 36 \log 3 \cdot \log 2$$

$$\log x = \frac{36}{6} \log 2$$

$$\log x = 6 \log 2$$

$$\log x = \log(2^6)$$

$$\log x = \log 64$$

$$x = 64$$

$$17) \log_2(x+2) + \log_2(x+3) = 1$$

$$\Rightarrow \log_2((x+2)(x+3)) = 1$$

$$\Rightarrow \log_2(x^2 + 3x + 2x + 6) = 1$$

$$\Rightarrow \log_2(x^2 + 5x + 6) = 1$$

$$\Rightarrow x^2 + 5x + 6 = 2^1$$

$$\Rightarrow x^2 + 5x + 6 - 2 = 0 \Rightarrow x^2 + 5x + 4 = 0$$

$$\Rightarrow x^2 + 5x + 4 = 0$$

$$\Rightarrow x^2 + 4x + x + 4 = 0$$

$$x(x+4) + 1(x+4) = 0$$

$$(x+1)(x+4) = 0$$

$$x = -1 \quad | \quad x = -4$$

Domain Check

For logarithm argument > 0 sign over

- $x+2 > 0 \Rightarrow x > -2$
- $x+3 > 0 \Rightarrow x > -3$

Together Condition $x > -2$

Check Solution

if $x = -4$: not valid Step 1's $x+2 = -2$

$-2 > 0$ is ~~negative~~ $\rightarrow$ log ~~not~~ undefined

if $x = -1$: valid Step 1's $x+2 = 1 > 0$ and

$$x+3 = 2 > 0$$

So that $x = -1$ is the Right answer

$$18) \quad 2J. (x+5) + 2J (x-7) = 2J 28$$

$\Rightarrow$ Step-1

$$\cancel{2J} ((x+5)(x-7)) = \cancel{2J} 28$$

$$\therefore (x+5)(x-7) = 28$$

$$\therefore x(x-7) + 5(x-7) = 28$$

$$\therefore x^2 - 7x + 5x - 35 = 28$$

$$\therefore x^2 - 2x - 35 - 28 = 0$$

$$\therefore x^2 - 2x - 63 = 0$$

$$a=1, b=-2, c=-63$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sqrt{b^2 - 4ac} = \sqrt{(-2)^2 - 4(1)(-63)}$$

$$= \sqrt{4 + 252}$$

$$= \sqrt{256}$$

$$= 16$$

$$x = \frac{-(-2) \pm 16}{2(1)} = \frac{2 \pm 16}{2}$$

$$x = \frac{2+16}{2} = 9$$

$$x = \frac{2-16}{2} = -7$$

but

by Condition (Argument > 0)
 $\therefore$ So $x=9$ is valid value &.